

L'Oréal x Monash Datathon

Theme: **Trendspotter**

Presented by **AL GLOW-RITHMS**

Our Team



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Problem: Noisy Review Data

How can L'Oréal turn **noisy, multilingual social media reviews** (e.g., YouTube, Facebook) into **early signals of beauty trends** and **reliable customer insights**?



66%

of people report seeing **spam** comments **regularly**



30%

of **comments** on brand ads are **negative (toxic)**

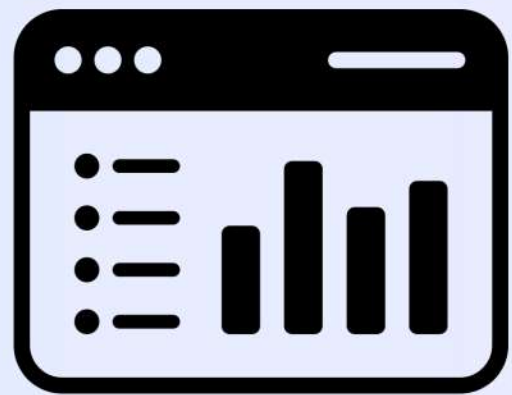


47%

of reviews on major websites are **suspicious (fake?)**

(Reference the Orbit Media Studios, Brandbastion and Capital One Shopping Research)

What the product is?



Dashboard

Sentiment trend analysis
(Customer Mood Tracker)



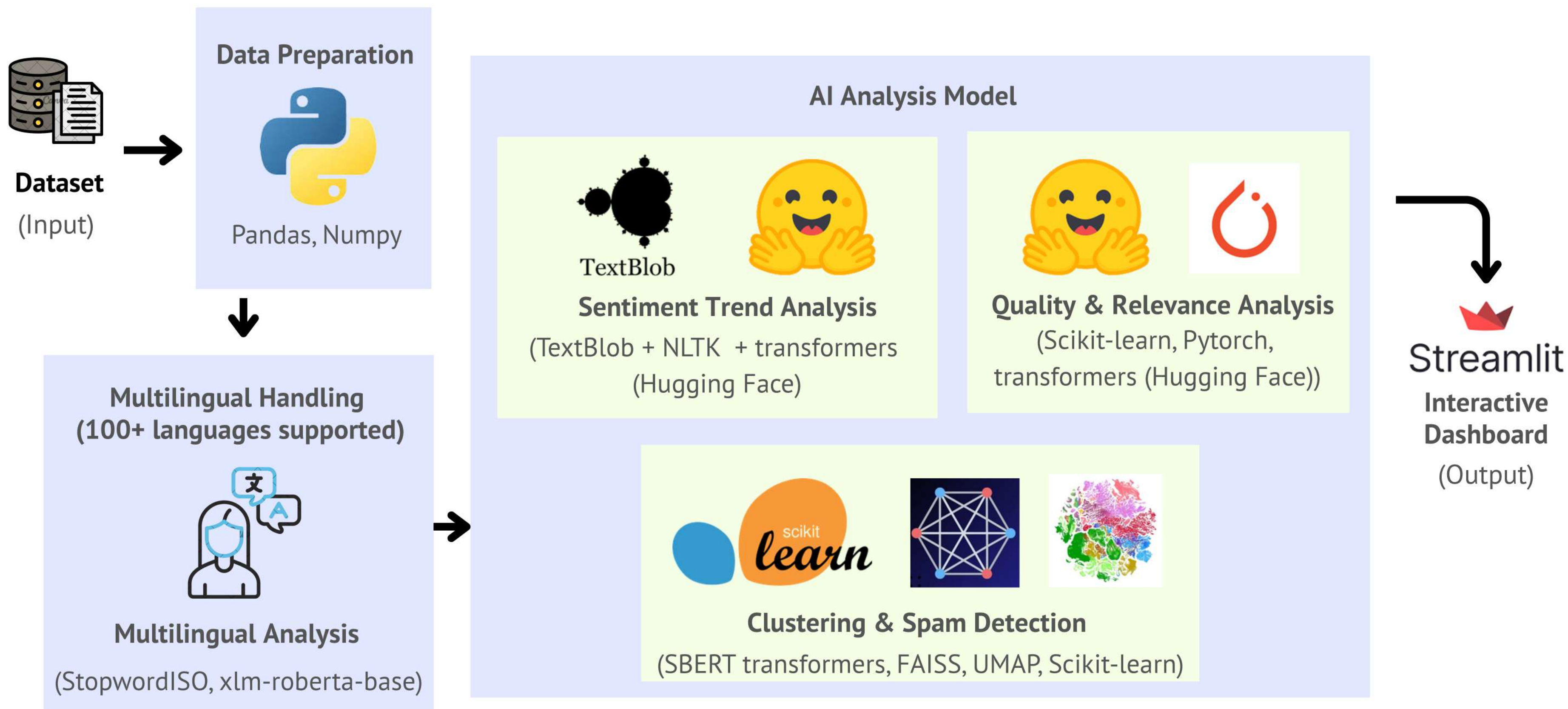
Clustering analysis
(Theme Discovery)



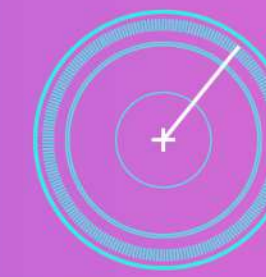
Quality & relevance analysis
(Quality Lens)



System Architecture



Application 1 | Customer Mood Tracker



Track

Highlight

Compare

Sentiment Trends Over Time

Filter categories (clusters):

Sunscreen / Skin... x

Time grouping:

Monthly

Key Sentiment Insights

😊 Total Positive

36,589

😞 Total Negative

44,043

😐 Total Neutral

23,361

📊 Positive %

35.2%

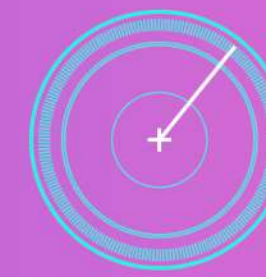
📊 Avg Sentiment

↓ 7.2%



Are customers getting happier?

Application 1 | Customer Mood Tracker



Are customers getting happier?

Key Sentiment Insights

 Total Positive

36,589

 Total Negative

44,043

 Total Neutral

23,361

 Positive %

35.2%

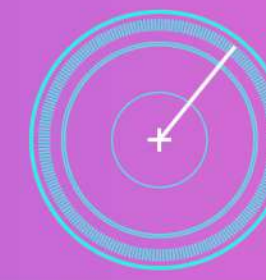
 Avg Sentiment

↓ 7.2%

AI Generated Business Insights

- Sentiment dropped 7.2% – people feel worse about the product
- More negatives than positives – 44k vs 36k comments
- Only 35.2% positive sentiment – not great performance
- Lots of neutral comments – many users are unsure
- Focus: Sunscreen/Skincare category needs attention

Application 1 | Customer Mood Tracker



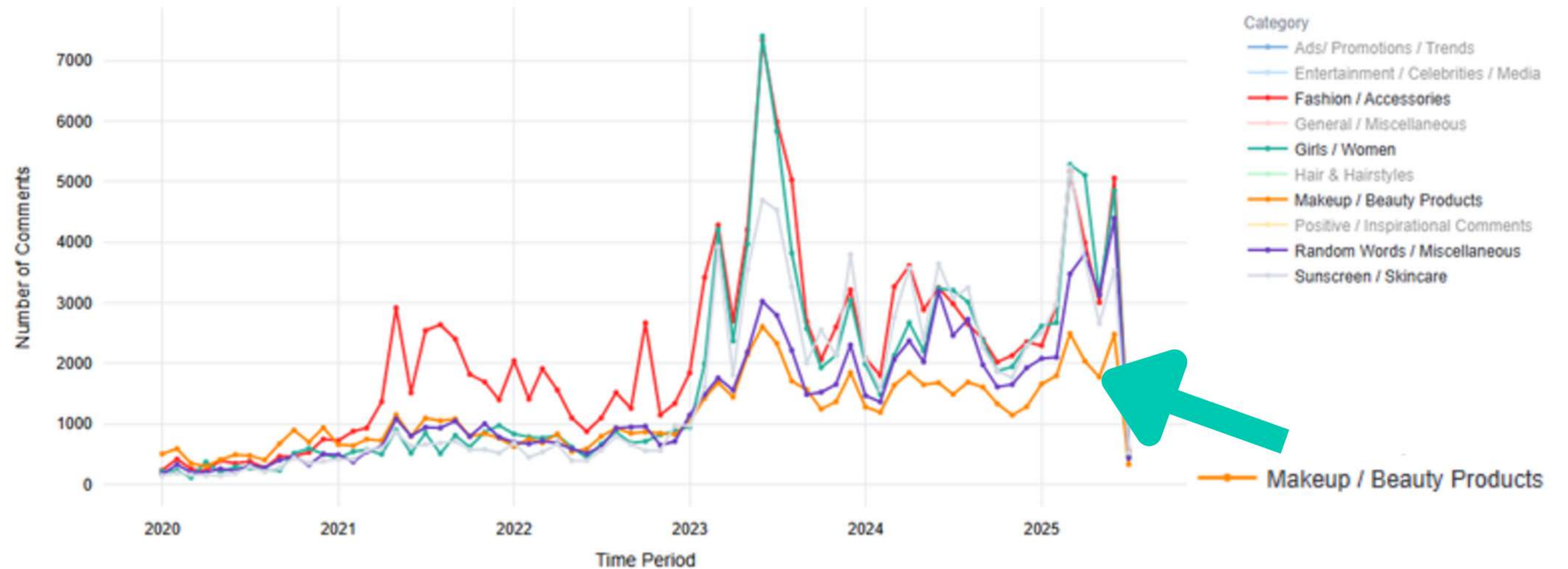
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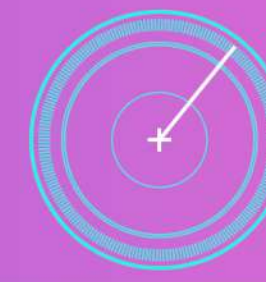
Top Categories Breakdown

Monthly Trends by Top 10 Categories

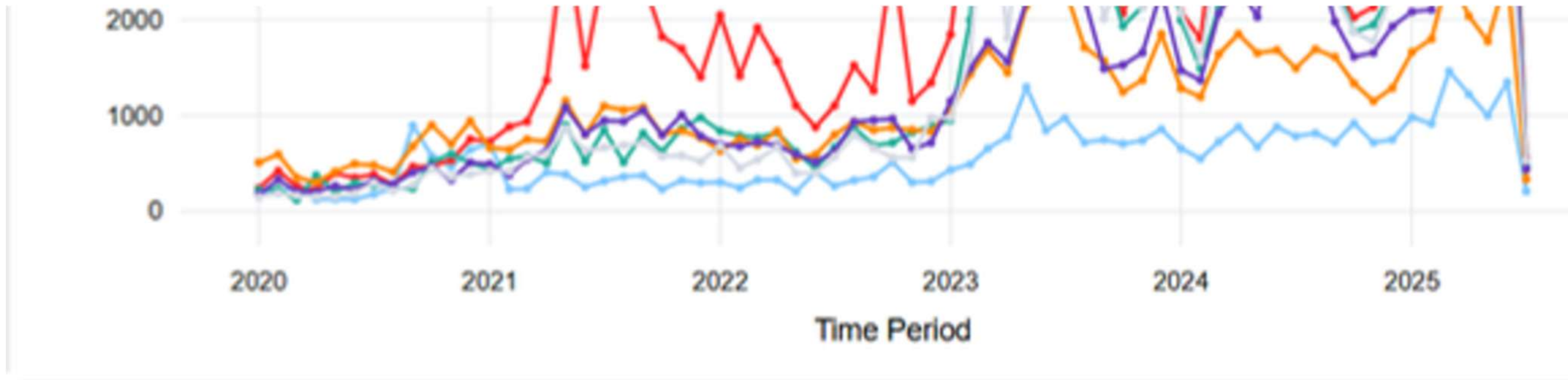


What's hot – and what's not?

Application 1 | Customer Mood Tracker



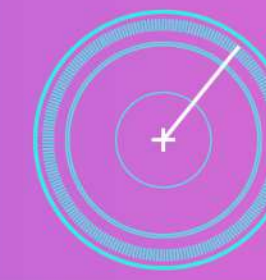
What's hot – and what's not?



AI Generated Business Insights

- Makeup/Beauty is dropping – big recent fall in mentions (orange line).
- General + Girls/Women spiked – these categories had a sharp rise.
- Sunscreen/Skincare is low – not getting much attention now.
- Fashion & Hair steady – consistent interest over time.

Application 1 | Customer Mood Tracker



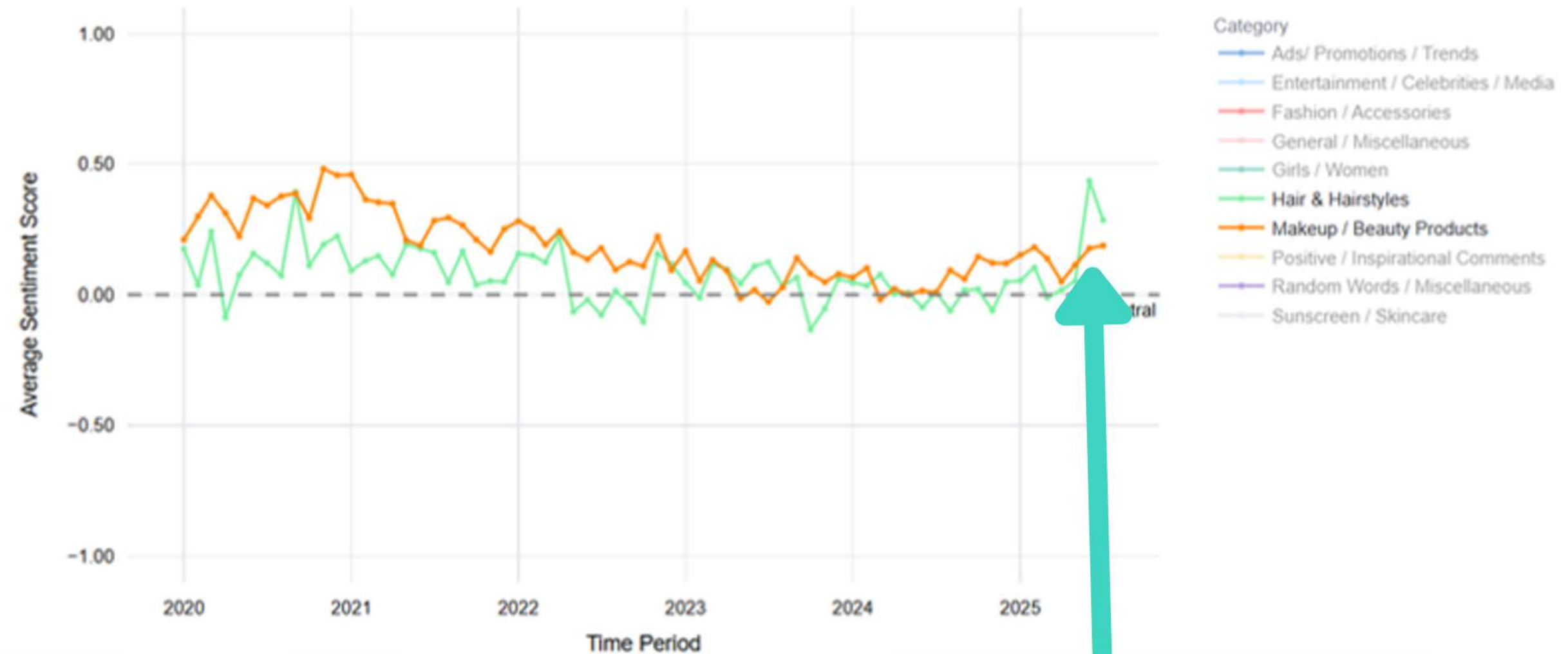
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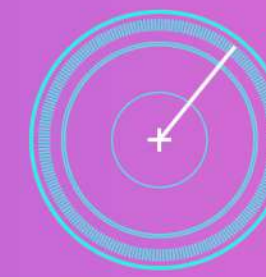
Average Sentiment by Category

Monthly Average Sentiment Score by Category (Top 10)

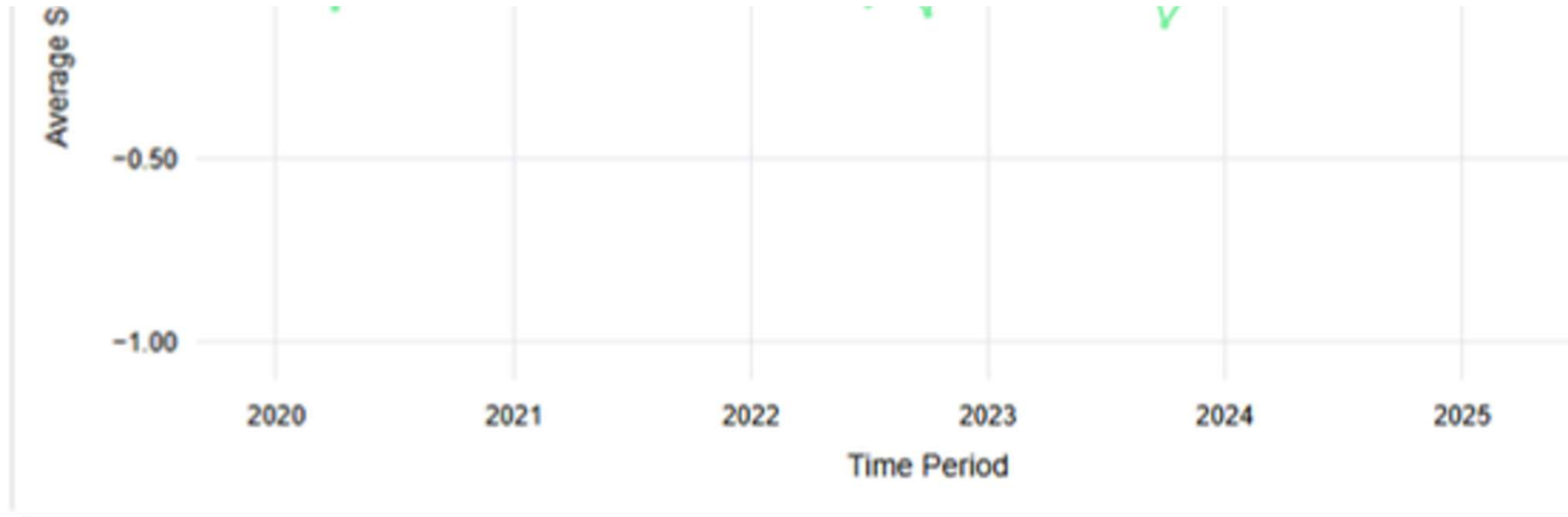


Which category do customers love more?

Application 1 | Customer Mood Tracker



Which category do customers love more?



AI Generated Business Insights

- Hair & Hairstyles has a higher sentiment than Makeup / Beauty in 2025
- Both are trending up, but Hair shows a sharper rise
- Hair is currently more favored by audiences
- Focus more on Hair content/products for better engagement

Application 2 | Theme Discovery

Track

Highlight

Compare

Top 10 Words per Cluster

Cluster	Top 10 Words
Fantasy / Princesses / Mythical	mermaid, witchcraft, princesa, princesses, linda, hair, witches, cleopatra, disney, princesse
Fashion / Accessories	makeup, women, hair, wear, indian, video, natural, aap, korean, cute
Food & Snacks	popcorn, rains, lollipop, ruined, sauce, seats, barbecue, phase, brother, hair
Foreign Words / Short Expressions	knigin, gzelsin, bleibst, paula, makeup, olmu, mrite, gzel, presque, abla
Gender / Feminism / Identity	handsome, sexy, genetics, hair, mujeres, bro, transgender, testosterone, mulheres, cute
General / Miscellaneous	cute, bro, aap, eyes, time, real, didi, bhi, kya, day
Girls / Women	girl, girls, woman, makeup, hair, boy, people, women, black, skin
Hair & Hairstyles	hair, natural, curly, ginger, blonde, cut, straight, hairstyle, black, color
Hair / Beauty / Cosmetic Random	wig, hair, wigs, makeup, pretty, wear, girl, natural, straighten, gorgeous
Haircare / Beauty Ingredients	acrylic, dip, skin, powder, oil, mask, nails, nail, castor, hydra

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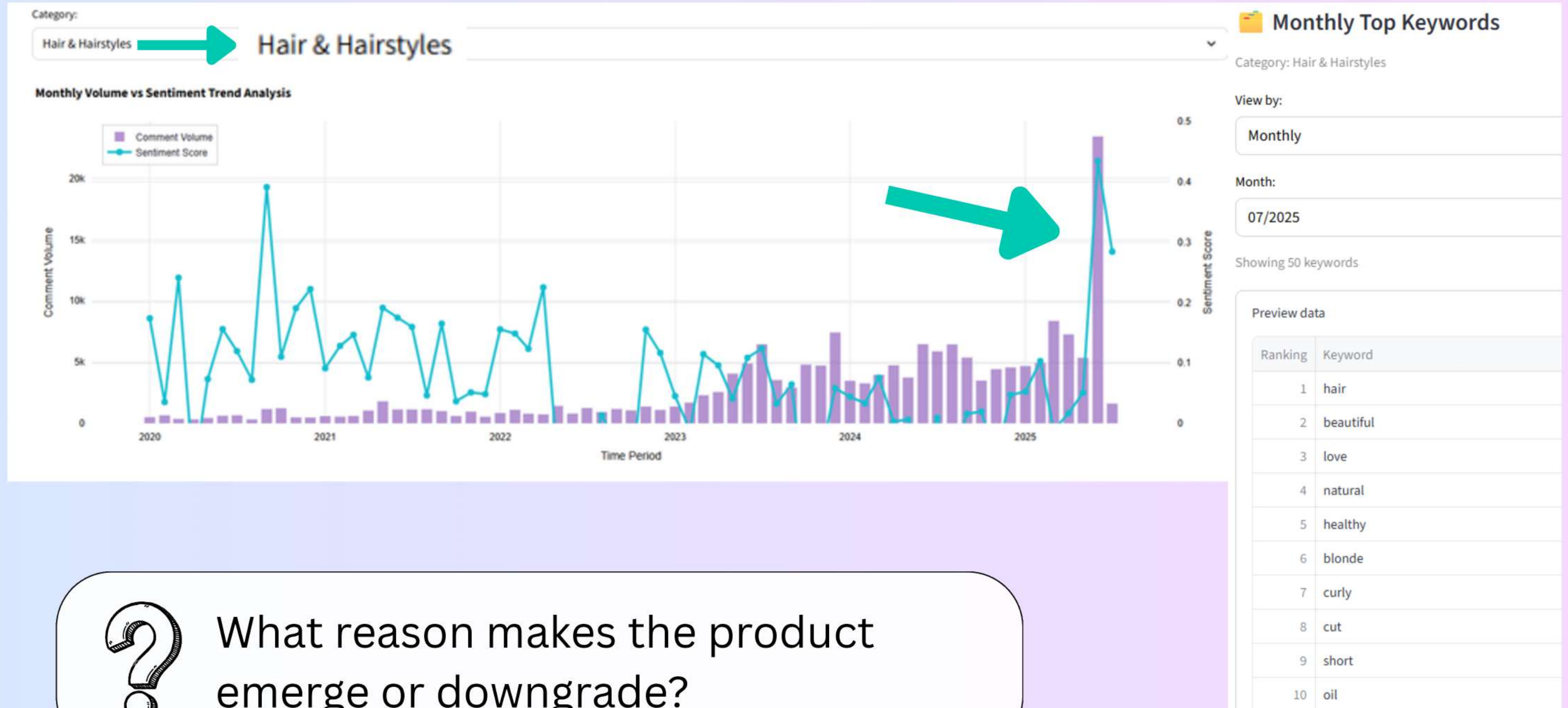
How to make the product attractive?

Application 2 | Theme Discovery

Track

Highlight

Compare



What reason makes the product emerge or downgrade?

Application 2 | Theme Discovery

Track

Highlight

Compare

VS YouTube Video Comparison

Compare sentiment analysis, comment volume, and spam detection metrics between two YouTube videos. Enter YouTube URLs below to analyze and compare their performance.

Video A

Link A

YouTube URL A:

<https://www.youtube.com/watch?v=ppnrOzGhGN0&t=6s>

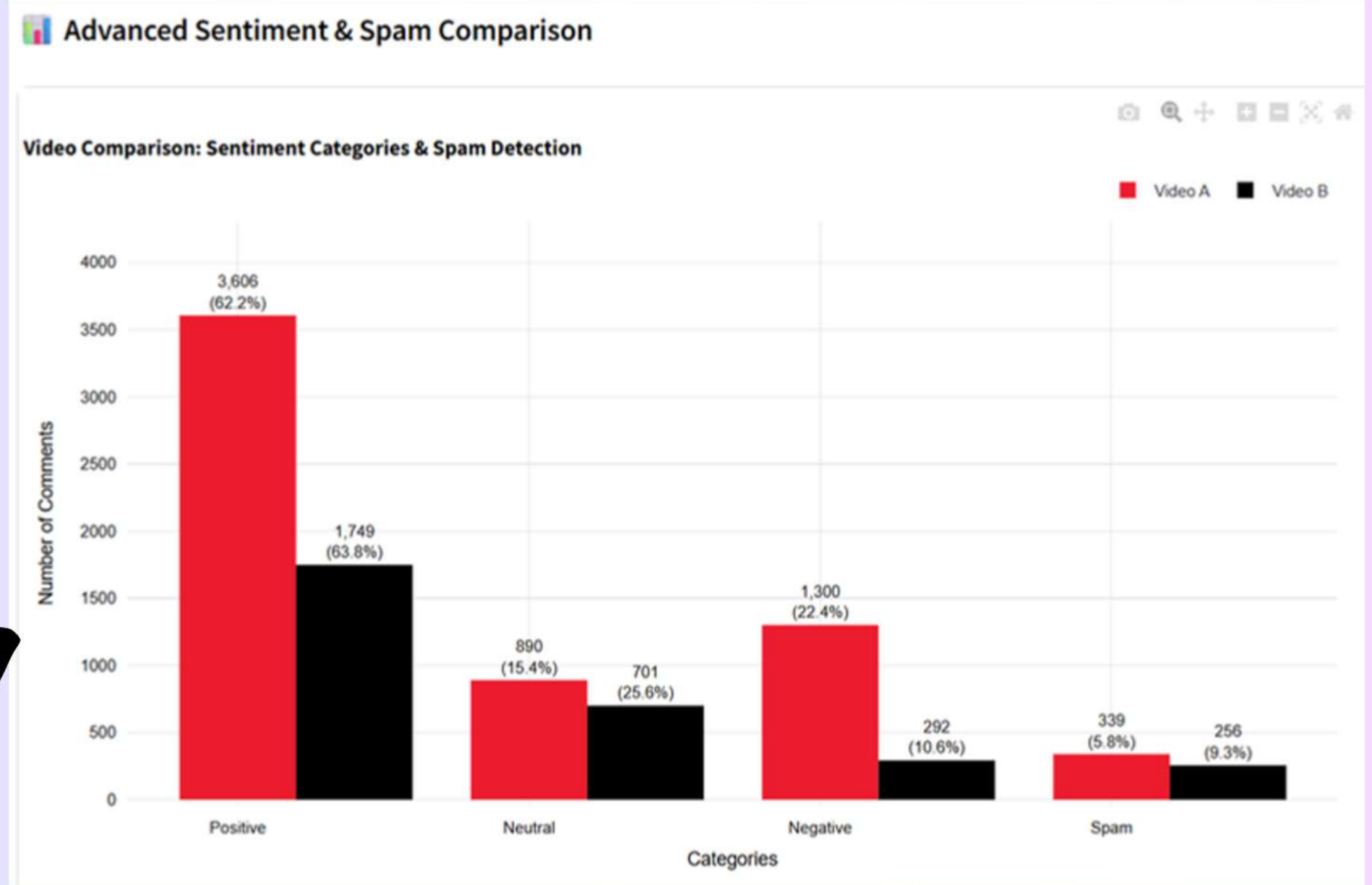
Video B

Link B

YouTube URL B:

<https://www.youtube.com/watch?v=example2>

Press Enter to apply



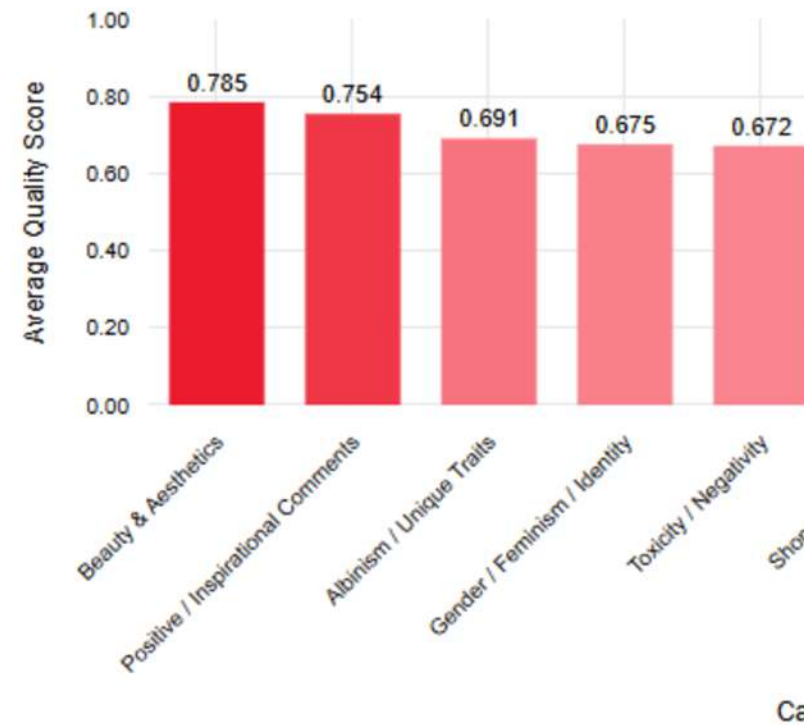
Is the product favourable?

Application 3 | Quality Lens

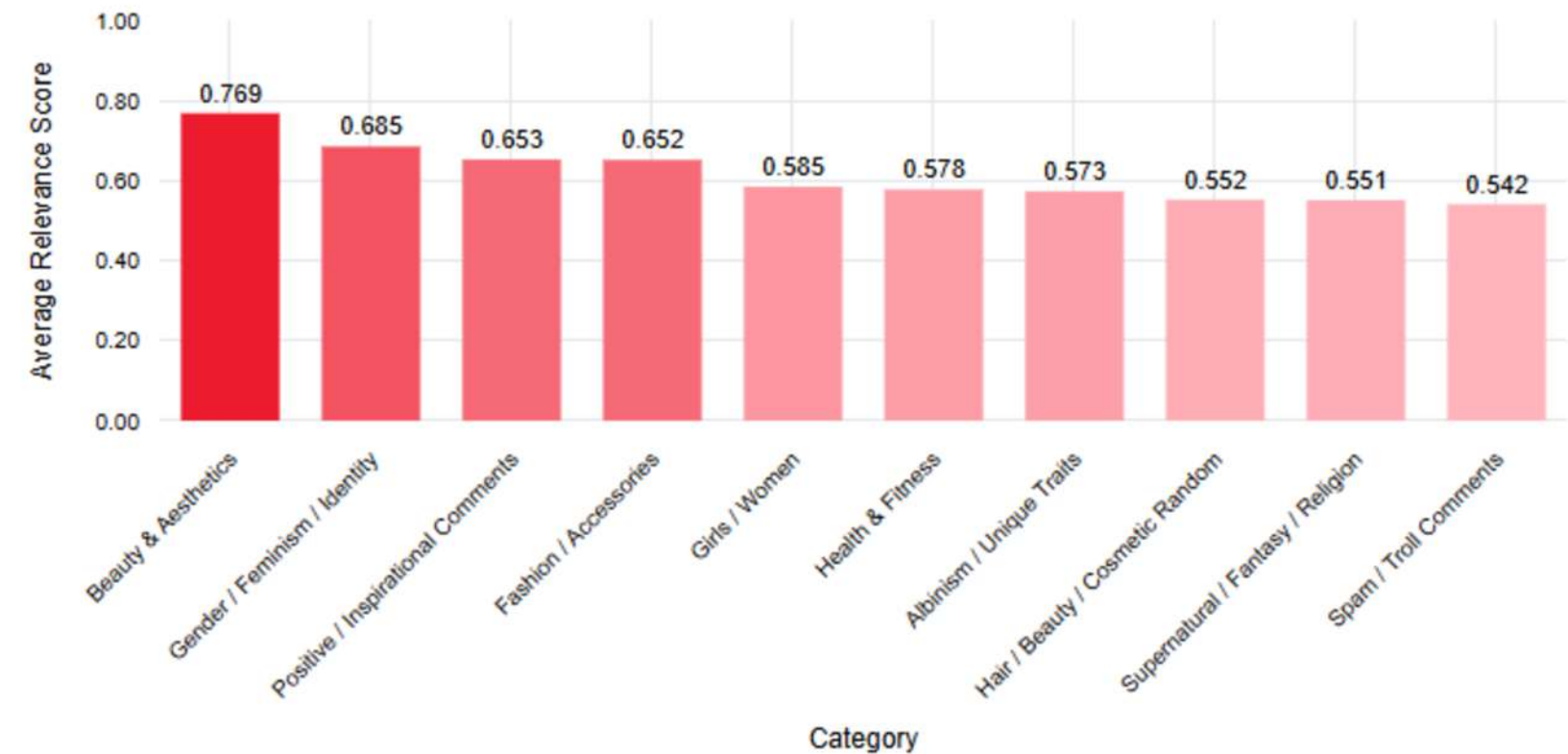
Track

Highlight

Top 10 Categories by Quality Score



Top 10 Categories by Relevance Score



Are the comments reliable or just noise?

Application 3 | Quality Lens

Track

Highlight

Spam Comments Analysis

Filter by Topic Category for Spam Analysis

All

Displaying 116464 spam comments for topic: 'All'

textOriginal	topicCategories_clean
I agree	['Lifestyle_(sociology)', 'Physical_attractiveness']
lol yeah	['Lifestyle_(sociology)', 'Physical_attractiveness']
I want to be like u but i am a boy	['Lifestyle_(sociology)', 'Physical_attractiveness']
@@kaitlynkong6701 I mean, thats also Mexican as well but yeah.	['Lifestyle_(sociology)', 'Physical_attractiveness']
I agree I do this everyday my hair is exactly like you but I have black hair	['Lifestyle_(sociology)', 'Physical_attractiveness']



Are the spam really spam?

Economic Feasibility



Expenses

\$0 licence fees (Open-source Stack)

~\$300–500/month

cloud storage and **compute cost** (for 2M+ comments per week)

~\$1K–2K/month

real-time analysis (AWS, Kafka, Spark)

~\$150/month

30K Business Insights from Gen-AI



Benefits/Revenue

Save ~\$20k+/year

by reduced **manual** comment review

30–40% less manual work

human only needed for **refinement**, not bulk tagging

Scale by choice

adjust dataset **size (%)**

10% data (2M) save \$800 compare to full data (20M)
per month

(Reference Indeed Malaysia and New York Times)

saves ~\$20k+ per year

compared to manual analysis

Competitive analysis

Feature	Our Solution	Brandwatch 	Bazaarvoice 	Chattermill 
Custom taxonomy + thresholds	✓ Yes ▾	Limited ▾	Limited ▾	Limited ▾
Multilingual + code-mix	✓ Yes ▾	Limited ▾	Limited ▾	Limited ▾
Spam/irrelevance filter	✓ Yes ▾	Limited ▾	Limited ▾	Limited ▾
Aspect sentiment + attributes	✓ Yes ▾	Limited ▾	Limited ▾	✓ Yes ▾
Quality/relevance scoring	✓ Yes ▾	✗ No ▾	Limited ▾	Limited ▾
Retrainable/ownable models	✓ Yes ▾	✗ No ▾	✗ No ▾	✗ No ▾

Conclusion

